Amrita Vishwa Vidyapeetham
Amrita School of Computing, Amaravati
Department of Computer Science and Engineering

23CSE113 User Interface Design 2-0-2-3

2024 – 2025 Even Semester

B. Tech CSE (2024-2028 Batch) **Semester: 2**

Academic year: 2024-25

Semester: Even

Department: Computer Science and Engineering

Programme: B.Tech. – CSE – II Semester

Course Code: 23CSE113

Course Type: Regular

Course Title: User Interface Design

Course Instructor(s) / Highlight Mentor:

Mr. Thangam S	Section A & Section B
Dr. D.Madhusudhana Rao	Section C

Credits(L-T-P-C): 2-0-2-3

Student Strength: CSE A - 69, CSE B - 69 & CSE C – 69 **Total:** 207

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(2) COs and CO-PO mapping

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(1) Syllabus with LTPC pattern.

23CSE113 User Interface Design L-T-P-C: 2-0-2-3

Syllabus

Unit 1

Introduction to Internet and Web design – Working of Web – Roles of Front-end, Back-end, and Full stack development – Web Server – Internet protocols – Web Hosting – HTML – HTML Tags – Create a simple Web Page – Marking up Text – Adding Links – Adding Images – SVG – Table Markup – Embedded Media – Web Based Forms – HTML Forms -CSS for Presentation – Basic Style Sheet- A CSS Style Primer – Using Style Classes – Using Style IDs – Formatting Text – Advance Typography with CSS – Working with Margins, Padding, Alignment and Floating, Responsive Web design, Introduction to version control systems.

Unit 2

CSS Box Model and Positioning – Creating Layouts Using Modern CSS- Backgrounds and Borders – CSS Transformation and Transitions – Animating with CSS and the Canvas – Dynamic Web Pages – Web Scripting – JavaScript programming Web Applications – Including JavaScript in HTML – HTML5 Form Controls and Validations – Document Object Model – DOM manipulation and events – Event Handlers.

Unit 3

Overview of UI Frameworks – User Interface Design Concepts in Web Development – Accessibility – Introduce to Electron – Overview of Electron – Setting up an Electron project – Creating a basic desktop application with Electron – Object communication – Introduction to other popular frameworks and libraries (e.g. JavaFX, WPF, GTK).

Textbook(s)

- Jennifer Kyrin, Julie Meloni, “Sams Teach Yourself HTML, CSS, and JavaScript All in One,” Third Edition, Pearson Education, inc., 2019.

Reference(s)

- Jennifer Niederst Robbin, “LEARNING WEB DESIGN A VEGINEER’S GUIDE TO HTML, CSS, JAVASCRIPT, AND WEB GRAPHICS,” Fifth Edition, O’Really Media Inc., 2018.

- Jessica Minnick, “Responsive Web Design with HTML 5 and CSS”, 9th Edition, Cengage Learning, Inc.,2021.
- Electron JS.
  https://www.electronjs.org/docs/lates
  https://www.electronforge.io/
  https://www.electronjs.org/fiddle

(2) COs and CO-PO mapping

Course Objectives

- The course focuses on the basic understanding of user interface design by applying HTML, CSS, and JavaScript. The course introduces the necessary skills to develop web applications with simple animation and transitions. An overview of the UI Framework is given. Working on the Internet is discussed. The protocols required are introduced.

Course Outcomes

CO1: To understand the basics and workings of the World Wide Web.

CO2: To understand the fundamentals of HTML 5.

CO3: To understand the fundamentals of CSS and JavaScript.

CO4: To design and deploy a simple web application.

CO-PO Mapping

PO/PSO / CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	2			2	2						3	2
CO2	2	2	2							3			3	2
CO3	2	2	2					1		3			3	2
CO4	2	2	3	2	3			3	3				3	2

(3) Course Plan with Module-Wise Objectives

Lect. Hrs	Module/Topic	Keywords	Objectives	COs	PO / PSO*
UNIT-1 Introduction to Internet and Web Design					
L1	Introduction to Internet and Web Design	Internet, Web Design	Understand the fundamentals of web design and how the web functions	CO1	PO1-3, 6,7/ PSO1,PSO2
L2	Working of Web	Client-Server Model, HTTP, HTTPS	Explain how web browsers and servers interact	CO1	
L3	Roles of Front-end, Back-end, and Full Stack Development	Front-end, Back-end, Full Stack	Identify different roles in web development	CO1	
L4	Web Server	Apache, Nginx, IIS	Learn about web servers and their role in hosting websites	CO1	
L5	Internet Protocols	HTTP, FTP, TCP/IP	Understand the protocols governing Internet communication	CO1	
L6	Web Hosting	Domain, DNS, Hosting Types	Learn how websites are hosted and made accessible	CO1	
L7	HTML and HTML Tags	HTML5, Elements, Structure	Introduce HTML syntax and elements for webpage creation	CO2	PO1-3, 10 / PSO,1PSO2
L8	Creating a Simple Web Page	HTML, Structure	Build a basic webpage using HTML	CO2	
L9	Marking Up Text	Paragraphs, Headings, Lists	Format text using HTML markup	CO2	
L10	Adding Links and Images	Anchor Tags, Image Tags	Learn how to add links and images to web pages	CO2	
L11	SVG and Embedded Media	Scalable Graphics, Audio, Video	Understand how to embed scalable graphics and media files	CO2	
L12	Table Markup	Tables, Rows, Columns	Create and format tables in HTML	CO2	
L13	Web-Based Forms	Input Fields, Form Elements	Implement user input forms in HTML	CO2	
L14	CSS for Presentation	CSS Basics, Selectors	Apply CSS styles to improve webpage aesthetics	CO3	PO1-3, 8,10 & PSO1,PSO2
L15	CSS Layouts and Styling	Margins, Padding, Alignment	Control the layout and positioning of elements	CO3	

L16	Responsive Web Design	Media Queries, Flexbox, Grid	Design responsive layouts for different screen sizes	CO3	
L17	Introduction to Version Control Systems	Git, GitHub	Learn the basics of version control in development	CO4	
	UNIT-2 CSS Box Model and Positioning				
L18	CSS Box Model and Positioning	Box Model, Positioning	Understand element spacing and positioning in CSS	CO3	
L19	Creating Layouts Using Modern CSS	Flexbox, Grid	Develop modern, flexible web layouts	CO3	
L20	Backgrounds and Borders	CSS Styling	Apply backgrounds and borders effectively	CO3	
L21	CSS Transformations and Transitions	Animation, Effects	Enhance web design with animations	CO3	
L22	Animating with CSS and Canvas	CSS Keyframes, HTML5 Canvas	Implement animations for dynamic effects	CO3	
L23	Dynamic Web Pages	DOM, JavaScript	Create interactive webpages	CO3	
L24	Web Scripting and JavaScript	Variables, Functions	Introduce JavaScript for interactivity	CO3	
L25	Including JavaScript in HTML	Script Tag, External JS	Implement JavaScript in web pages	CO3	
L26 – L27	HTML5 Form Controls and Validations	Form Validation, Input Types	Validate user input using HTML5	CO3	
L28	Document Object Model (DOM)	Nodes, Tree Structure	Learn DOM manipulation for interactivity	CO3	
L29 – L30	Event Handlers	Click Events, Mouse Events	Use JavaScript events for user interactions	CO3	
	UNIT-3 Overview of UI Frameworks				
L31	Overview of UI Frameworks	Bootstrap, Tailwind, Material UI	Introduce common UI frameworks	CO4	PO1-5, 8,9 & PSO1,PSO2
L32 – L34	User Interface Design Concepts in Web Development	UX/UI, Design Principles	Learn core UI/UX principles	CO4	
L35 – L36	Accessibility in Web Development	WCAG, ARIA	Understand accessibility standards	CO4	
L37	Introduction to Electron	Desktop Apps, Web Tech	Explore Electron for building desktop apps	CO4	

L38 – L39	Overview of Electron	Node.js, Chromium	Understand Electron's architecture	CO4	
L40	Setting Up an Electron Project	Installation, Configuration	Learn to set up an Electron project	CO4	
L41 – L42	Creating a Basic Desktop Application with an Electron	UI, App Lifecycle	Develop a simple desktop app	CO4	
L43 - L44	Object Communication in Electron	IPC, Main & Renderer Process	Handle inter-process communication	CO4	
L45	Introduction to Other Popular Frameworks and Libraries	JavaFX, WPF, GTK	Compare various UI frameworks	CO4	

(4) Evaluation Pattern: 70-30

Assessment	Internal	End Semester
Midterm	20	
*Continuous Assessment (Theory)	10	
*Continuous Assessment (Lab)	40	
**End Semester		30 (50 Marks; 2 hours exam)

*CAT includes Quizzes and Tutorials
*CAL – Can be Lab Assignments, Projects, Case Studies, and Reports.
**End Semester – Lab-based examination

Component Name	Weightage	Exam Name	Exam Marks
Mid-1	20	M-1	50
CAT (Theory)	10	Assignment-1 Assignment-2	10 10
CAL (Lab)	40	Lab Assessment Lab Report	30 10
End Semester	30	End Semester Theory Examination	50

List of Experiments:

1. **Create a simple webpage using basic HTML elements.**
2. **Designing a Multi-Page Website** – Link multiple HTML pages together to form a basic website.
3. A**pplying CSS for Text Styling** – Use CSS to change font style, size, color, and text alignment.
4. **CSS Box Model and Layout** – Experiment with margins, padding, and borders in CSS.
5. **Creating Responsive Web Design** – Use media queries to adjust content for different screen sizes.
6. **Using CSS Flexbox and Grid** – Develop a webpage layout using modern CSS techniques.
7. **Creating a Themed Webpage** – Design a simple themed webpage (e.g., portfolio, event page) using CSS styling techniques.
8. **Creating a Multi-Page Website with Navigation** – Design a website with a navigation bar and internal linking (HTML &CSS).
9. **Designing a Web Form with Validation** – Use CSS & HTML5 constraints for form validation
10. **Introduction to JavaScript Variables and Data Types** – Create a script demonstrating variable usage.
11. **JavaScript Functions and Event Handling** – Implement event-driven programming with onclick, mouseover, etc.
12. **Form Validation using JavaScript** – Validate user inputs (e.g., email, password, number fields) using JavaScript.
13. **JavaScript DOM Manipulation** – Change webpage content dynamically using the DOM.
14. **Fetching Data from an API using JavaScript** – Use AJAX & Fetch API to display real-time data.
15. **Building a Simple Web Application** – Develop a To-Do List, Calculator, or Weather App using JavaScript.